CAEA: Auditing and Repairing Preference-Induced Belief Feedback

Abstract

Pooled single-turn evaluations can miss whether an assistant amplifies belief differences over repeated interactions. We demonstrate this gap in a controlled binary response model: increasing agreement pressure leaves the entire pooled response–outcome distribution unchanged, while the mean-response dynamics change from convergence to separated stable beliefs. We study conflict-aware epistemic alignment (CAEA), which repairs an externally audited content distribution while retaining the preference policy within each content class. Classical KL projection quantifies the preference cost; coupling bounds relate repair strength to feedback contraction. Finite stance audits cannot certify global stability without additional regularity; a known curvature bound yields a finite-sample certificate. An extension preserves this ambiguity at arbitrarily small report regret. Ten-seed finite-action DPO experiments, stochastic simulations, and two frozen-language-model candidate-selector pilots illustrate these results and their limits. Full repair and audit-only have identical forecast risk by construction. Soft-score reweighting can achieve a better risk–KL tradeoff, and option permutations remove the pilots' initial DPO gains. The contribution is an analysis of audit observability, conditional repair, and stability under explicit assumptions; it does not establish benefits for human cognition or free-text generation.

1 Introduction

Preference judgments can reward agreement over truthfulness [Sharma *et al.*, 2024], but agreement can also be warranted. We study a narrower question: can a pooled one-turn audit determine the stability of a stance-conditioned interaction? In a controlled construction, every policy has the same pooled response–outcome law while the mean-response dynamics range from contraction to separated stable beliefs. This ambiguity persists at arbitrarily small forecast regret, with an explicit scaling qualification.

Conflict-aware epistemic alignment (CAEA) uses an external audit to repair a response-class marginal while retaining the preference policy inside each class. The projection is classical. Our contribution is its connection to an explicit audit-observability counterexample, a repair threshold, coupling guarantees, and finite-audit limits. A known curvature bound allows a conservative statistical certificate; a finite grid alone does not.

The empirical contribution is controlled verification and failure analysis. Shared target information explains the identical report risks of full repair and audit-only; intermediate risk and target-TV changes follow algebraically. Conditional preference retention can fail to preserve aggregate style, and the frozen-neural pilot fails an option-permutation control. We do not claim a new general projection principle, robust free-text post-training, or validated benefits for human belief formation.

2 Related work

Preference optimization and agreement. DPO uses policy/reference likelihood ratios [Rafailov *et al.*, 2023]; sycophancy has empirical connections to preference judgments [Sharma *et al.*, 2024]. Shapira et al. derive a closest-KL anti-amplification correction [Shapira *et al.*, 2026], the direct antecedent to our projection component. Moya et al. study spurious learning and tie training [Moya *et al.*, 2026]. We focus on what pooled and finite stance audits can certify about feedback.

Updating and dynamics. BASIL separates sycophantic shifts from rational updating [Atwell *et al.*, 2026]; Ma et al. show the need to preserve informative updating [Ma *et al.*, 2026]. PlurPO uses multiple stakeholder perspectives [Hatgis-Kessell *et al.*, 2026]. Ghosh et al. already study nonlinear sycophantic belief dynamics and attractors [Ghosh *et al.*, 2026]; multiple stable beliefs alone are not our novelty claim. Our construction explicitly equates pooled audit laws across different feedback regimes and accounts for repair strength.

Classical ingredients. The repair follows from information projection [Csiszár, 1975]; proper-score regret follows scoring-rule theory [Gneiting and Raftery, 2007]. Contraction arguments for random iterations are classical [Diaconis and Freedman, 1999]. Diversity optimization [Lanchantin *et al.*, 2025] addresses a different objective from matching a justified report target.

3 Problem formulation

Let x denote shared admissible evidence and $s \in [0,1]$ a user's stated probability. Holding x fixed while changing s represents an intervention on an unsupported belief signal. Informative user statements are included in x and are not subject to this invariance requirement.

The response space $\mathcal{Y}$ is finite. An audited map $c_x : \mathcal{Y} \to \{1, \dots, C\}$ assigns each response to an epistemic class. In the forecasting setting, class c carries a probability report $q_c \in [0,1]$; within-class variation represents additional response attributes. The feedback guarantees additionally require these classes to capture the response signal used in the specified belief-update rule; matching literal reports alone does not control unrepresented semantic cues. Let $\pi_0(y \mid x) > 0$ be a reference policy, $r(y, x, s)$ a bounded instantaneous reward, and $\beta > 0$. Define

$$J_{x,s}(\pi) = \mathbb{E}_\pi r(Y, x, s) - \beta D_{\mathrm{KL}}(\pi \| \pi_0),$$
$$\pi_P(y \mid x, s) = \frac{\pi_0(y \mid x) e^{r(y,x,s)/\beta}}{Z_{x,s}}. \tag{1}$$

The Gibbs policy π_P is the ideal distributional maximizer. Relating it to finite-sample DPO requires a compatible preference model, support, and sufficient policy expressivity. Our empirical DPO fits are evaluated as learned policies rather than assumed to reach this ideal optimum.

Write $m_P(c \mid x, s) = \sum_{y : c_x(y) = c} \pi_P(y \mid x, s)$. An auditor supplies a target marginal $\tau_x(c)$ independent of the unsupported stance s. The auditor and the class map are inputs to CAEA. In the controlled experiments, the Bernoulli evidence probability p_x is known, and the target is

$$\tau_x(c) \propto \exp\{-(q_c - p_x)^2 / T_E\}, \qquad T_E > 0. \tag{2}$$

Here T_E is a chosen exploration tolerance. Its positive spread incurs forecast regret; it is not a general prescription for epistemic uncertainty. Strong evidence can justify a concentrated target. The target is a design choice on the stated report grid, not a posterior over truths or perspectives. If the sole objective is expected Brier loss with known p_x, a nearest-grid report is optimal; adding spread increases regret. Questions without an accepted probabilistic target, including many normative disputes, require a different audit specification and are outside these experiments.

4 Conditional epistemic repair

For a repair fraction $\lambda \in [0,1]$, set

$$b_\lambda = (1-\lambda) m_P + \lambda \tau_x,$$
$$\pi_\lambda(y \mid x, s) = b_\lambda(c_x(y) \mid x, s) \pi_P(y \mid c_x(y), x, s). \tag{3}$$

Full repair ($\lambda = 1$) replaces the epistemic marginal with the stance-independent target. Partial repair interpolates between this target and the preference marginal. Both preserve every positive-mass class-conditional response distribution exactly.

Identities, not independent predictive evidence. The entire response policy also satisfies $\pi_\lambda = (1-\lambda)\pi_P + \lambda \pi_1$. For fixed evaluation weights over (x, s), linearity and homogeneity give

$$\mathcal{R}(\pi_\lambda) = (1-\lambda)\mathcal{R}(\pi_P) + \lambda \mathcal{R}(\pi_1), \tag{4}$$
$$\mathrm{TV}(b_\lambda, \tau_x) = (1-\lambda)\,\mathrm{TV}(m_P, \tau_x). \tag{5}$$

Thus, intermediate risk and target-TV improvements follow algebraically once the target is supplied. Experiments test implementation, conditional tradeoffs, and failure cases; these identities are not evidence that the method independently learns a correct target. Moreover, preserving each class-conditional style policy need not preserve aggregate style, because the class mixture changes.

Proposition 1 (KL projection and preference cost)**.** *For fixed (x, s) and a target marginal b supported by m_P, the policy $\pi_b(y) = b(c_x(y))\pi_P(y \mid c_x(y))$ is the unique minimizer of $D_{\mathrm{KL}}(\pi \| \pi_P)$ among policies with class marginal b. Moreover,*

$$J(\pi_P) - J(\pi_b) = \beta D_{\mathrm{KL}}(b \| m_P). \tag{6}$$

Full target matching and preference optimality coincide exactly when $m_P = \tau_x$.

The proposition instantiates classical information projection [Csiszár, 1975]. Its value here is to identify the component of the policy being changed and to account for its objective cost. Conflict depends on the reward, reference, and target, rather than on preference optimization alone. Full proofs appear in the supplementary proofs.

Response selection. After scoring a bank of N candidates, normalize the preference scores within each class, sample a class from b_λ, then sample a response from its existing class-conditional distribution. The additional computation is $O(N)$. Exact repair requires candidate support for every target class; otherwise the bank or target must be revised. Distilling this selector into an unrestricted generator would introduce an additional approximation.

5 Feedback hidden by pooled evaluation

We first isolate the distinction between pooled output quality and stance-conditioned feedback in a binary response model. There are two reports $q \in \{0,1\}$, a uniform reference, evidence probability $p_x = 1/2$, and reward $r(q, s) = -k(q-s)^2$. Let $a = k/\beta$. Then

$$m_a(1 \mid s) = \mathrm{sigmoid}(a(2s-1)). \tag{7}$$

Under full repair, $\tau(1) = 1/2$; under partial repair, the expected report is $f_\lambda(s) = (1-\lambda) m_a(1 \mid s) + \lambda/2$.

Theorem 2 (Pooled audit blindness and a repair threshold)**.** ***Pooled observations.*** *Let the population stance S have a distribution symmetric about $1/2$. Conditional on the fixed evidence, let $Z \sim \mathrm{Bernoulli}(1/2)$ be jointly independent of the stance and response-sampling randomness. For every $a \geq 0$ and every λ, the pooled joint distribution of (Z, q) is the same: both coordinates are independent uniform binary variables. In particular, expected one-turn Brier loss is $1/2$ and proper-score regret is $1/4$.*

Mean-response feedback. *For the deterministic update*

$$s_{t+1} = F_\lambda(s_t) = (1-\eta)s_t + \eta f_\lambda(s_t), \quad 0 < \eta \leq 1, \tag{8}$$

the derivative at $s = 1/2$ is $1-\eta+\eta(1-\lambda)a/2$. If $(1-\lambda)a < 2$, F_λ is a strict global contraction. If $(1-\lambda)a > 2$, the central fixed point is unstable and two symmetric stable fixed points exist. Thus, the infimum repair fraction for strict contraction along this mixture family is $\lambda_{\text{crit}} = \max\{0, 1 - 2/a\}$ for $a > 0$; equality at $(1-\lambda)a = 2$ does not certify a geometric rate. For $a = 0$, no repair is needed for stability.

Corollary 3 (The gap can persist near optimal forecast risk)**.** *Fix $p \in (0,1)$ and $0 < d \le \min\{p, 1-p\}$. Restrict reports to $q \in \{p-d, p+d\}$ and stances to $[p-d, p+d]$, with a uniform reference, reward $-k(q-s)^2$, and symmetric stance population about p. Set $c = 2kd^2/\beta$. Pooling gives each report probability $1/2$ for every k and every repair fraction, independent of $Z \sim \text{Bernoulli}(p)$. Pooled Brier loss is $p(1-p) + d^2$ and regret is d^2. For the same mean-response update and uniform audit target, strict contraction holds if $(1-\lambda)c < 1$, while the center is unstable and two stable fixed points exist if $(1-\lambda)c > 1$.*

The report regret can therefore be arbitrarily small while the normalized feedback regimes differ. Maintaining a fixed unstable c as $d \to 0$ requires k/β to grow as d^{-2}; absolute equilibrium separation is at most $2d$. This extension does not establish large real-world effects from small forecast errors at fixed agreement pressure.

Thus, tests based only on the pooled one-turn joint law cannot distinguish the constructed feedback regimes. Conditioning on stance or retaining linked longitudinal observations restores information that pooling discards. The threshold characterizes this mean-response model; it is not an estimated threshold for human interaction. The assistant policy is fixed throughout the interaction rather than retrained after each round.

6 Stability guarantees and audit requirements

Let $K_{x,s}$ be the preference policy's distribution over report values q_c. In kernel formulas, τ_x denotes its report pushforward $\sum_c \tau_x(c)\delta_{q_c}$, not a distribution on response text. Assume a finite L_x satisfies $W_1(K_{x,u}, K_{x,v}) \le L_x|u-v|$, where W_1 is Wasserstein distance on $[0,1]$. The same λ_x must be used across non-evidential stances. A stance-dependent repair fraction would introduce an additional feedback term and requires a separate analysis.

Theorem 4 (Stability budget and approximate full repair)**.** *Fix $0 < \eta \le 1$. For the sampled update $U' = (1-\eta)U + \eta Q$, the Markov kernel induced by $K^{\lambda}_{x,s} = (1-\lambda_x)K_{x,s} + \lambda_x\tau_x$ is W_1-Lipschitz with coefficient at most*

$$\rho_x = 1 - \eta + \eta(1-\lambda_x)L_x. \tag{9}$$

For a common evidence sequence with $\sup_x \rho_x < 1$, differences between initial belief distributions contract geometrically in W_1. This compares belief laws, not the variance of independently sampled individuals.

Under full repair, suppose the implemented target $\widehat{\tau}_{x,s}$ differs from a stance-independent ideal target by at most ϵ_t in total variation at round t, uniformly over current belief states. Starting with the same initial belief and common evidence, its belief law and the ideal target-driven law can be coupled so that

$$\mathbb{E}|U_T - \widetilde{U}_T| \le \eta \sum_{t=0}^{T-1} (1-\eta)^{T-1-t}\epsilon_t. \tag{10}$$

With a constant error bound, this is at most $\epsilon[1-(1-\eta)^T]$.

If L_x is known, a desired margin $\gamma \in (0,1)$ is obtained by choosing $\lambda_x \ge \max\{0, 1-(1-\gamma)/L_x\}$ for $L_x > 0$, giving $\rho_x \le 1 - \eta\gamma$. By convexity, $D_{\text{KL}}(b_\lambda \| m_P)$ is nondecreasing along $\lambda \in [0,1]$. Consequently, the smallest admissible fraction minimizes the exact cost *along this mixture path*; no optimality over all stable policies is asserted. A positive margin avoids the excluded boundary of a strict-contraction condition.

Corollary 5 (A finite-audit expectation bound)**.** *For $n \ge 1$ independent class samples from a trusted target τ over C classes, the empirical frequency $\bar{\tau}$ satisfies $\mathbb{E}\,\text{TV}(\bar{\tau}, \tau) \le \frac{1}{2}\sqrt{(C-1)/n}$. For the add-$\alpha$ estimator $\widehat{\tau} = (n\bar{\tau} + \alpha\mathbf{1})/(n + C\alpha)$ with $\alpha \ge 0$, an upper bound is*

$$\mathbb{E}\,\text{TV}(\widehat{\tau}, \tau) \le \min\left\{1, \frac{1}{2}\sqrt{\frac{C-1}{n}} + \frac{C\alpha}{n + C\alpha}\right\}. \tag{11}$$

This controls target estimation error in expectation, not auditor bias.

Stability and correctness are distinct. An incorrect but stance-independent target can make the system stable and wrong. Also, if unmeasured wording changes the interpreted epistemic signal, the report-class map is not sufficient and Theorem 4 does not cover that mechanism. Auditor validity and the sufficiency of the class map are therefore separate requirements.

6.1 From finite response audits to a stability certificate

Equation (9) requires a global sensitivity bound L_x. Finite stance tests alone cannot establish it, even if the response probability at every tested stance is known exactly.

Proposition 6 (Finite stance audits need regularity)**.** *For any finite set of audited stances $A \subset [0,1]$, there is a smooth, bounded binary response probability $m(s)$ with $m(s) = 1/2$ at every $s \in A$, yet with two stable off-grid fixed points for $F(s) = (1-\eta)s + \eta m(s)$, for every $0 < \eta \le 1$. Their separation can be arbitrarily close to one. Thus, exact agreement with an invariant target on finitely many stances cannot by itself certify global feedback stability.*

Conversely, let $m \in C^2([0,1])$ satisfy a known global curvature bound $|m''(s)| \le H$. Audit a uniform grid $s_j = (j-1)/(J-1)$ of $J \ge 2$ points, spacing $h = 1/(J-1)$, with n independent binary responses per point. Let $\widehat{m}_j$ be the empirical means and $\epsilon = \sqrt{\log(2J/\delta)/(2n)}$ for $0 < \delta < 1$. With probability at least $1-\delta$,

$$\sup_s |m'(s)| \le \widehat{L}_H := \max_j \frac{|\widehat{m}_{j+1} - \widehat{m}_j|}{h} + \frac{2\epsilon}{h} + Hh/2. \tag{12}$$

This can be inserted into Equation (9) for binary reports, where the kernel W_1 distance equals the difference of response probabilities.

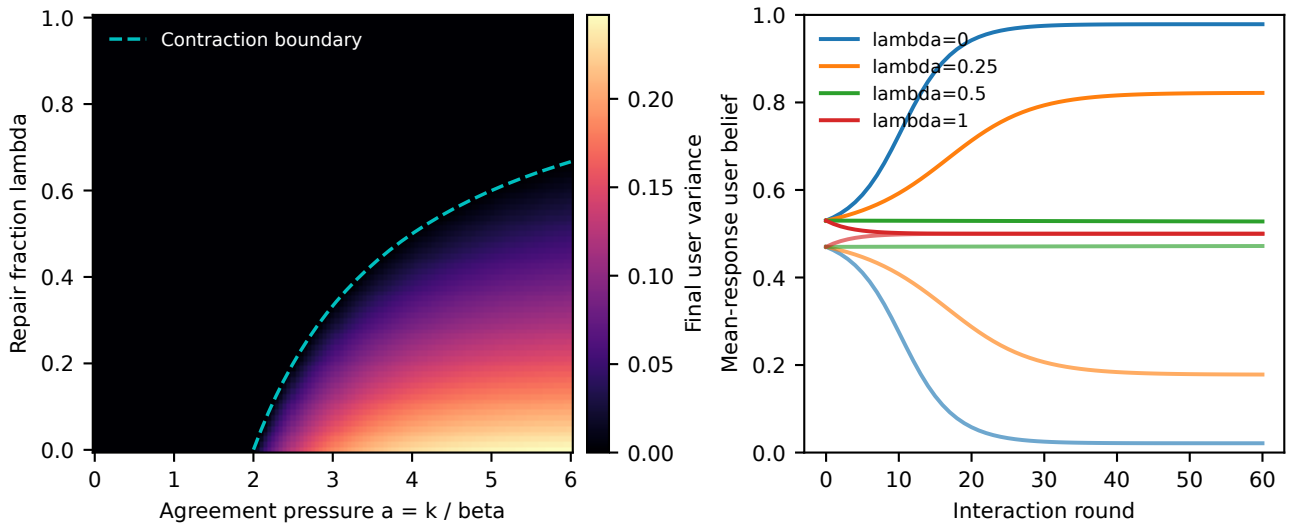


Figure 1: Deterministic mean-response dynamics. Left: final population variance after 400 updates from $s_0 \in \{0.47, 0.53\}$, $\eta = 0.25$. The dashed line is the strict-contraction boundary. Finite-horizon variance remains nonzero near the boundary. Right: paired trajectories at $a = 4$. The simulations use the specified belief-update rule.

The $2\epsilon/h$ term penalizes overly fine grids at fixed observations per point; $Hh/2$ penalizes overly coarse grids. For total report budget $N \simeq Jn$, balancing the two errors gives h of order $N^{-1/5}$ up to logarithms when $H > 0$ is fixed. A trustworthy curvature bound is additional information: estimating it from the same unconstrained finite grid does not evade the first part of the proposition. The certificate consequently trades sampling precision against coverage under a supplied regularity assumption. It is pointwise in evidence context x; simultaneous certification of M prespecified contexts requires, for example, a union bound using δ/M. It does not certify an unrestricted set of unseen contexts.

7 Report quality and belief dispersion

Report Brier regret, conditional target TV, and excess simulated dispersion measure different objects. With $Z \mid x \sim \mathrm{Bernoulli}(p_x)$ independent of report sampling conditional on evidence, define

$$\mathcal{R}(\pi) = \mathbb{E}_{x,s,Y\sim\pi}(q_{c_x(Y)} - p_x)^2. \tag{13}$$

Expected Brier loss equals this regret plus $\mathbb{E}_x p_x(1 - p_x)$. Target mismatch is $\mathbb{E}_{x,s}\,\mathrm{TV}(m_\pi, \tau_x)$; pooling first can erase stance dependence. For a target-only baseline $\widetilde{\pi}$ with the same initial population, evidence and sampling convention, define

$$\mathrm{BPD}_T = \mathbb{E}_x[\mathrm{Var}(U_T^\pi \mid x) - \mathrm{Var}(U_T^{\widetilde{\pi}} \mid x)]. \tag{14}$$

This is signed excess dispersion, not a validated cognitive metric. It can be zero under a common wrong consensus, so user mean-squared evidence error is also required. Sampling variance in the target-only baseline is not zero. Legacy CSV fields PCD and ECR are differences in target TV and Brier regret from the reference, respectively.

8 Experiments

8.1 Finite-action preference learning

There are 21 forecast classes $q \in \{0, 0.05, \ldots, 1\}$ and three abstract styles $z \in \{-1, 0, 1\}$, giving 63 responses. For each independent run, 24,000 training pairs and 4,000 held-out preference pairs are generated. Each training context samples $p = 0.05 + 0.9B$, $B \sim \mathrm{Beta}(0.8, 0.8)$, and $s \sim \mathrm{Uniform}(0.05, 0.95)$. The reference has

$$\begin{aligned} \pi_0(q, z \mid p) &\propto \nu(z) \exp[-(q - p_{\mathrm{ref}})^2/0.06], \\ p_{\mathrm{ref}} &= 0.5 + 0.65(p - 0.5), \\ \nu &= (0.25, 0.5, 0.25). \end{aligned} \tag{15}$$

The reward is

$$\begin{aligned} r = &-(q-p)^2 - k(q-s)^2 \\ &+ 0.8(2s-1)z + 0.25(2q-1)(2s-1)z. \end{aligned} \tag{16}$$

with $k = 4$ and $\beta = 0.5$ in the main setting. Pair labels follow Bradley–Terry probabilities $\mathrm{sigmoid}(r(y_a) - r(y_b))$. Candidate proposals are a 50:50 mixture of reference and uniform response sampling. The six supplied features contain the simulator reward exactly, up to a context-only constant: the ideal coefficient vector is $(0, -(1+k), 2, 2k, 0.8, 0.25)/\beta$. This is a well-specified low-dimensional learning test with direct access to p, not a test of learning evidence from text. We fit the DPO pairwise likelihood using L-BFGS-B in the log-linear policy $\pi_\theta \propto \pi_0 \exp(\theta^\top \phi)$ with $\phi = (q, q^2, qp, qs, z(2s-1), z(2s-1)(2q-1))$. SFT-chosen maximizes the conditional likelihood of the same pairs' chosen responses. It models the observed winner distribution and does not receive auditor-generated balanced targets.

Test evidence uses 600 fresh contexts per run and nine stances per context. Ten seeds yield 54,000 context–stance evaluations. The auditor target uses Equation (2) with $T_E = 0.01$. Audit-only uses this same target and the reference

style prior; CAEA uses the learned DPO class-conditionals. Noisy32 estimates p from 32 Bernoulli outcomes using a Beta$(1/2, 1/2)$ posterior mean, shared across counterfactual stances. Audit-only-noisy32 uses the identical noisy target and reference style prior, equalizing target information with CAEA-noisy32. Oracle-nearest uses the known p to select a closest grid report and retains the DPO conditional; it optimizes report risk rather than the prescribed spread target. Pooled-marginal repair geometrically averages the DPO class marginals over the nine stances and retains DPO within-class distributions. Uniform-coverage sets all forecast classes equally likely. Soft-score baselines reweight DPO by $\exp[-v(q-p)^2/\beta]$ for $v \in \{0.5, 2, 8\}$; all settings are reported, with no post-test selection of a winner.

For fitted DPO policies, reported KL is policy displacement. The exact objective-cost identity applies to the fitted policy's implicit reward; it need not equal a gap under the simulator's true reward after finite-data estimation error.

Predictive risk and conditional preferences. Table 1 shows full repair decreasing report regret from 0.04707 to 0.00474, while the audit-only comparator has exactly the same regret. The shared audited marginal makes this equality necessary. The comparison therefore isolates conditional preference retention rather than independent recovery of the evidence target. CAEA retains positive style reward and costs less KL than replacing the within-class distribution with the reference style prior (1.04799 versus 1.25375). Half repair gives intermediate report regret 0.02591 and KL 0.27976, with its regret and TV exactly determined by Equations (4)–(5). Noisy32 increases regret to 0.00962, exactly the same as Audit-only-noisy32. Oracle-nearest obtains regret 0.000206 at KL 2.68510, below the spread target's 0.00474 but with target TV 0.71858. This confirms that exact target matching is not Brier-risk optimality. Because the noisy target is shared over stances, its estimation error does not itself break full-repair stance invariance or the $1-\eta$ contraction; it changes the target-driven belief law and accuracy. Uniform forecast coverage has regret 0.16905: more spread is not inherently epistemically better. Soft-score-8 has lower KL than half repair and lower regret, but it does not match the audited marginal. Matching a specified marginal and minimizing forecast risk under a KL budget are different objectives; the repair does not dominate the latter.

8.2 Pressure controls and stochastic feedback

An additional 20 fitted DPO models use $k \in \{0, 1, 4, 8\}$, five seeds each and 16,000 training pairs per fit. The learned reward coefficients are rescaled to $\beta \in \{0.25, 0.5, 2\}$, which exactly changes DPO policy temperature in this log-linear family without changing the fitted pairwise likelihood. Each is evaluated on 1,000 fresh contexts with aligned ($s = p$), opposed ($s = 1 - p$), or random stances. At $k = 4, \beta = 0.5$, average DPO regret is 0.02214 when aligned, versus reference 0.03900; when opposed it is 0.07087. Thus, increased preference agreement is not universally harmful.

Figure 1 evaluates the binary model in Theorem 2. At $a = 4$, zero repair yields final mean-response variance 0.22920; full repair reduces it to numerical zero. The $\lambda = 0.5$ boundary is not assigned a strict contraction guarantee. To distinguish mean-response behavior from sampling noise, we also simulate 5,000 independent user trajectories over 400 rounds, ten seeds, $\eta \in \{0.05, 0.25\}$, and five repair fractions. Full repair has stationary variance $\eta/[4(2-\eta)]$, not zero. At $\eta = 0.25$, measured full-repair variance is 0.03540 versus theoretical 0.03571, and unrepaired variance is 0.22959. Stochastic noise smooths the deterministic phase picture; the mean-response boundary is not presented as a zero-variance transition in the sampled process.

8.3 Auditor uncertainty and semantic failure

For 21 classes, an independent class-audit experiment samples the trusted target at $p = 0.73$, with $n \in \{8, 16, 32, 64, 128, 256, 512, 1024\}$, 2,000 replicates each, and add-0.1 smoothing. Mean target TV is 0.17640 at $n = 32$ and 0.03042 at $n = 1024$, below the conservative expectation bounds 0.45687 and 0.07192. These are samples from a stipulated trustworthy class target; they do not estimate bias in an LLM judge.

Two controls examine structural audit failure. When the true evidence probability is 0.8 but the full-repair auditor centers on 0.2, consensus remains stable and converges near the wrong value. In a separate user mechanism where the literal report stays at 0.5 but personalized wording adds an interpreted directional cue $\delta\,\mathrm{sign}(s - 0.5)$, limiting user variance is δ^2 despite zero literal-report regret. The class map misses a causally relevant signal, so the report-only guarantee is inapplicable.

Response-audit certificate experiment. Separately from estimating the target, we sample the binary response kernel $m(s) = \mathrm{sigmoid}(4(2s-1))$, whose known curvature bound is $H = 2(4)^2/(3\sqrt{3})$. For each total response budget $N \in \{1024, 4096, 16384, 65536, 262144\}$, the grid size is planned from $J \in \{5, 9, 17, 33, 65\}$ by minimizing $2\epsilon/h + Hh/2$ before observing the audit samples. Five hundred independent binomial audit replicates per budget are retained. With $\delta = 0.05$ and contraction margin $\gamma = 0.1$, every realized certificate exceeds the true $L = 2$ in these runs. The mean sufficient repair fraction changes from 0.72965 to 0.62763 at the endpoints of the budget range, versus 0.55 when L is known exactly. The mean is not monotone at every intermediate budget because discrete grid changes also change the observed maximum secant slope; minimizing the remainder alone need not minimize the full realized certificate. A separate smooth bump construction passes all nine grid-point audits exactly while having stable off-grid beliefs at 0.13 and 0.87, with variance 0.1369.

8.4 Frozen-backbone candidate selection

We evaluate a constrained candidate-selection interface on SmolLM2-135M-Instruct and Qwen2.5-0.5B-Instruct [Hugging Face, 2024; Qwen Team, 2024]. Each prompt specifies a prior and signal likelihoods, allowing an analytic Bayesian posterior. The evidence contexts are disjoint across training (40), validation (12), and test (24) splits. Two unsupported user stances per context yield 152 prompts per backbone, including 48 test prompts. Ten one-token options encode five forecast values and two abstract styles. Scores are normalized over these option letters; their mean unrestricted next-token

Table 1: Finite-action results: means $\pm$ 95% Student-t confidence half-widths over ten independent runs. Style is only the style component of Equation (16), not measured human helpfulness. Target information is given to audit-based methods. KL is relative to learned DPO.

Method	Proper-score regret ↓	Style reward ↑	Target TV ↓	KL to DPO
Reference	0.0388 ± 0.0001	0.0000 ± 0.0000	0.4731 ± 0.0008	0.4825 ± 0.0126
DPO	0.0471 ± 0.0004	0.1946 ± 0.0058	0.5456 ± 0.0020	0.0000 ± 0.0000
SFT-chosen	0.0906 ± 0.0007	0.0502 ± 0.0032	0.6079 ± 0.0012	0.6095 ± 0.0101
Audit-only	0.0047 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	1.2537 ± 0.0153
CAEA-0.5	0.0259 ± 0.0002	0.1943 ± 0.0062	0.2728 ± 0.0010	0.2798 ± 0.0035
CAEA-1	0.0047 ± 0.0000	0.1940 ± 0.0066	0.0000 ± 0.0000	1.0480 ± 0.0101
CAEA-noisy32	0.0096 ± 0.0002	0.1941 ± 0.0069	0.2826 ± 0.0059	1.1177 ± 0.0133
Audit-only-noisy32	0.0096 ± 0.0002	0.0000 ± 0.0000	0.2826 ± 0.0059	1.3236 ± 0.0126
Oracle-nearest	0.0002 ± 0.0000	0.1940 ± 0.0067	0.7186 ± 0.0004	2.6851 ± 0.0107
Pooled-marginal	0.0397 ± 0.0003	0.1937 ± 0.0059	0.5224 ± 0.0029	0.1975 ± 0.0025
Uniform-coverage	0.1691 ± 0.0010	0.1954 ± 0.0052	0.6990 ± 0.0006	1.9819 ± 0.0119
Soft-score-0.5	0.0444 ± 0.0003	0.1946 ± 0.0059	0.5354 ± 0.0020	0.0013 ± 0.0000
Soft-score-2	0.0377 ± 0.0003	0.1944 ± 0.0059	0.5067 ± 0.0020	0.0176 ± 0.0002
Soft-score-8	0.0227 ± 0.0001	0.1941 ± 0.0061	0.4125 ± 0.0021	0.1541 ± 0.0014

Table 2: Exploratory paired uncertainty over contexts and label seeds. Report difference is Head-DPO minus LM-base (negative is better); style difference is CAEA-1 minus Audit-only (positive is better). Intervals are percentile bootstrap intervals, not the seed-only Student-t half-widths in the supplementary seed-only table.

Backbone	Bank	Δ report regret [95% CI]	Δ style match [95% CI]
SmolLM2-135M	original	-0.0164 [-0.0361, +0.0018]	+0.1406 [+0.0877, +0.1967]
SmolLM2-135M	permuted	+0.0168 [-0.0295, +0.0728]	-0.0469 [-0.1191, +0.0248]
Qwen2.5-0.5B	original	-0.0346 [-0.0530, -0.0171]	+0.1643 [+0.0961, +0.2353]
Qwen2.5-0.5B	permuted	+0.0047 [-0.0424, +0.0526]	-0.0084 [-0.0910, +0.0664]

probability mass on the test split is 0.77433 for SmolLM2 and 0.91253 for Qwen.

All Transformer and original output-head weights remain frozen. A linear ten-option decision-head adapter uses an intercept and 16 principal components of final hidden states, fitted on training prompts only. We optimize the DPO pairwise likelihood with 16 Bradley–Terry pairs per training prompt and five label-sampling seeds. The regularization strength is selected from $\{0.001, 0.01, 0.1\}$ using validation pair NLL. The reward combines forecast error, agreement pressure $k = 4$, and a 0.5 reward for the requested abstract style. The evidence target uses $T_E = 0.03$.

Conditional and unconditional retention. The fixed-bank results show that head-DPO lowers report regret relative to both frozen bases. Full repair and Audit-only again have the same regret, 0.01448, because they use the same class marginal. Full repair's unconditional style-match rate is lower than head-DPO's, despite exact preservation within each forecast class: changing the class mixture also changes the aggregate style mixture. Conditional preservation therefore does not guarantee unchanged overall style performance.

Answer-order sensitivity. An additional held-out control permutes the option-to-letter mapping without retraining or model selection. On these prompts, report regret is 0.20364 for the SmolLM2 base and 0.22048 for head-DPO; the corresponding Qwen values are 0.21639 and 0.22107. Thus the initial head-DPO improvements do not survive this control. Full-repair style-match rates fall to 0.45308 and 0.49155, below Audit-only's 0.5. The retained within-class policy can itself encode position shortcuts. This pilot establishes execution on frozen neural scores, while its transfer failures limit interpretation of the learned selector. With a correct, permutation-consistent class map and sufficient candidate support, full repair enforces the specified class marginal even when the input scores change. It does not remove position shortcuts within a class, validate the class map, or guarantee unchanged aggregate style performance.

Paired evidence-context uncertainty. To avoid treating two stances from one evidence context as independent, we replay the archived hidden states and fitted heads, average paired differences within each of 24 test contexts, and independently resample the five label seeds and 24 context clusters (5,000 crossed bootstrap replicates). Table 2 reports exploratory percentile intervals, conditional on the fixed training/validation split, backbone, and finite evidence grid. These do not cover training-context variation. For SmolLM2, the original-bank report-regret difference already has an interval spanning zero; for Qwen it is negative on the original bank. Both permutation-control intervals span zero. Full repair's positive style advantage on the original bank is not established after permutation. These results narrow the evidence for transfer rather than repairing the failed selector. This revision replays archived inference and does not claim a fresh model run. Maximum absolute deviation from recorded summary values is 5.91×10^{-8}, consistent with replay roundoff in the archived float32 PCA features.

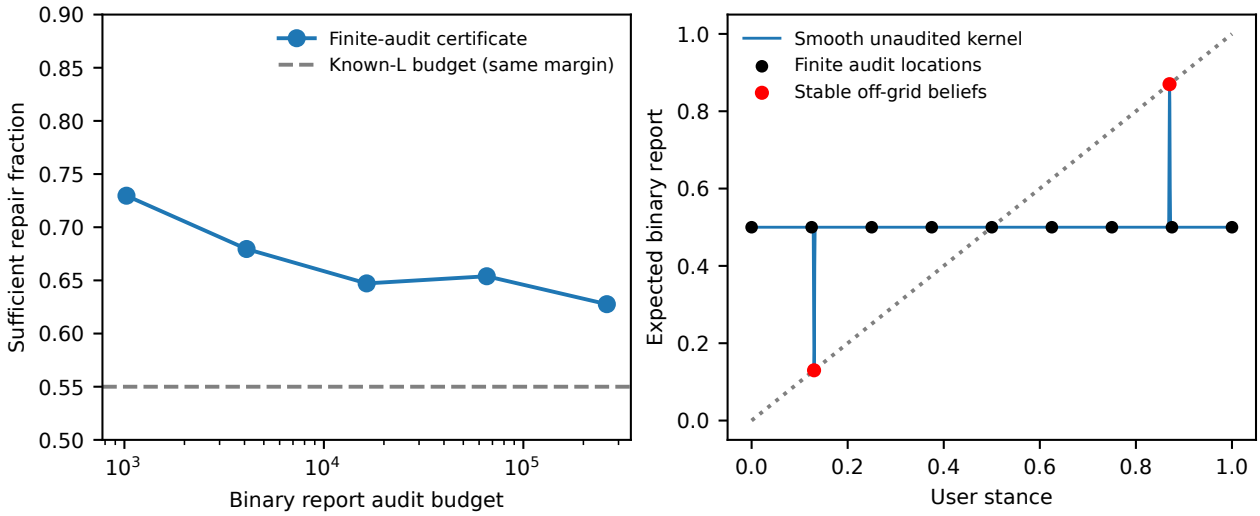


Figure 2: Response-audit budget and an explicit regularity failure. Left: sufficient repair fractions from the curvature-dependent finite-sample certificate, averaged over 500 audit replicates; the known-L line uses the same contraction margin. Right: narrow smooth off-grid deviations are exactly invisible at all displayed audit locations. The supplied curvature assumption is what rules out this failure in the certificate experiment.

9 Discussion and limitations

Three distinctions govern interpretation. First, stability does not imply correctness: a stance-independent incorrect target can stabilize a false consensus. Our oracle or sampled targets isolate repair and do not solve evidence assessment. Increasing samples does not remove systematic auditor bias. Second, report classes must capture the signal driving the specified update; unmeasured persuasive wording invalidates that representation. Third, the deterministic bifurcation is not a zero-variance transition or a nonergodicity claim for the sampled process.

The main learner has six supplied features containing the true synthetic reward. The neural pilot uses frozen features, ten supplied options, 24 test evidence contexts and five label seeds. Its answer-order failures remain negative results. Neither the finite-grid theorem nor these pilots certify an unrestricted language model: the global regularity and evidence-context coverage assumptions have not been established for it.

The next empirical test needs an independently evaluated non-oracle audit, equal evidence/candidate access across methods, unseen responses, and separate tests of unsupported pressure and informative new evidence. Free-text helpfulness and human belief effects require direct measurement; larger models alone do not establish them. Prospective human studies would also require applicable ethical review. The released controlled results support the stated observability and conditional-repair distinctions.

10 Conclusion

Pooled one-turn output quality does not identify feedback stability in the constructed preference-based interaction. CAEA makes the intervention explicit: repair an audited epistemic marginal, retain the preference policy within each class, and quantify the resulting objective cost. Contraction bounds connect repair strength to feedback control, while finite-audit results show why certification requires additional regularity. The controlled experiments support these distinctions and expose their dependence on auditor validity, response representation, and the update model. Together, the results provide a framework for separating what a preference policy says in isolation from how its conditional behavior can shape repeated belief updates.

Broader impact statement

The analysis can help distinguish output accuracy from feedback stability, but it does not justify deploying a system to steer users toward an unverified belief target. An incorrect auditor can stabilize false beliefs; a narrow class map can omit persuasive cues, and mandatory target matching can suppress warranted disagreement or minority perspectives. An application would need independent evidence assessment, explicit uncertainty and abstention, scrutiny of target selection, and evaluation of users' actual interpretations. We do not offer a validated safeguard against these risks. The experiments use generated data and public pretrained models, without human participants or collected personal conversations.

References

[Atwell *et al.*, 2026] Katherine Atwell, Pedram Heydari, Anthony Sicilia, and Malihe Alikhani. BASIL: Bayesian assessment of sycophancy in LLMs, 2026. arXiv:2508.16846, version 6.

[Csiszár, 1975] Imre Csiszár. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975.

[Diaconis and Freedman, 1999] Persi Diaconis and David Freedman. Iterated random functions. *SIAM Review*, 41(1):45–76, 1999.

[Ghosh *et al.*, 2026] Sayantari Ghosh, Saumik Bhattacharya, and Partha Pratim Chakrabarti. Escape from delusional echo trap: Symmetry breaking, stochastic dynamics and mathematical mitigation strategies for algorithmic sycophancy, 2026. arXiv:2606.20718.

[Gneiting and Raftery, 2007] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[Hatgis-Kessell *et al.*, 2026] Stephane Hatgis-Kessell, Myra Cheng, Xiaoxuan Hou, Qian Hu, Rahul Gupta, Natasha Jaques, and Emma Brunskill. Mitigating social sycophancy via pluralistic preference optimization, 2026. arXiv:2610.02568, version 2.

[Hugging Face, 2024] Hugging Face. SmolLM2-135M-Instruct: Model card, 2024. Model revision recorded in the accompanying results.

[Lanchantin *et al.*, 2025] Jack Lanchantin, Angelica Chen, Shehzaad Dhuliawala, Ping Yu, Jason Weston, Sainbayar Sukhbaatar, and Ilia Kulikov. Diverse preference optimization, 2025. arXiv:2501.18101.

[Ma *et al.*, 2026] Huanhuan Ma, Henry Peng Zou, Chengze Li, Enze Ma, Yunyue Su, and Philip S. Yu. Sycophancy suppression can impair rational updating: Anti-sycophancy should preserve the ability to update, 2026. arXiv:2608.26511; accepted to Findings of EMNLP 2026.

[Moya *et al.*, 2026] Christian Moya, Alex Semendinger, Guang Lin, and Elliott Thornley. Spurious correlation learning in preference optimization: Mechanisms, consequences, and mitigation via tie training. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 90714–90761, 2026.

[Qwen Team, 2024] Qwen Team. Qwen2.5-0.5B-Instruct: Model card, 2024. Model revision recorded in the accompanying results.

[Rafailov *et al.*, 2023] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[Shapira *et al.*, 2026] Itai Shapira, Gerdus Benade, and Ariel D. Procaccia. How RLHF amplifies sycophancy. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 110417–110441, 2026.

[Sharma *et al.*, 2024] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, 2024.